

Waffy Ahmed

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Education

Georgia Institute of Technology | Atlanta, GA

December 2024

Bachelor of Science in Computer Science

Faculty Honors, Dean's List, Zell Miller Scholar

Experience

Software Engineer

January 2025 – Present

The Home Depot

Atlanta, GA

- Increased resilience by deploying and validating Kubernetes Horizontal Pod Autoscaling (HPA) for a core service previously capped at 20 static pods, reducing mean latency from 121 to 72 ms and errors by 89% while increasing throughput from 24.9M to 34.8M reqs/week and enabling dynamic scaling between 50–100 pods in production.
- Reconstructed deployment workflows for legacy Java 1.8 Tomcat services supporting business-critical transaction systems lacking operational ownership, rebuilding CI/CD pipelines and enabling zero-downtime Cassandra credential rotation while eliminating a critical platform outage risk capable of halting transaction capture.
- Led testing and validation to disable a legacy service component handling ~27% of transaction volume (14.5 million daily captures/lookups) using BigQuery and Postman, supporting a phased retirement effort involving 12 teams.
- Reduced mean time to recovery (MTTR) ~50% by implementing an automated Kubernetes rolling-restart job for 20+ microservices, replacing slow Argo Rollouts and accelerating recovery from memory-leak/state-drift issues.
- Automated Spinnaker deployment pipelines for 15+ microservices with cdk8s, standardizing multi-environment release workflows using TypeScript, Terraform, and GCP, reducing deployment times by 25%.

Software Engineer Intern

May 2023 – July 2023, May 2024 – July 2024

The Home Depot

Atlanta, GA

- Developed an internal product information page for cashiers, streamlining self-checkout processes and reducing customer wait times for associate interventions by ~25%.
- Leveraged Java, React, and TypeScript to retrieve and surface validated product data across all self-checkout registers, mitigating customer theft by ~\$750,000 annually.
- Optimized the efficiency of configuring 40,000+ registers across 2,300 stores by 90%, in collaboration with 2 interns.
- Truncated maximum deployment time for register updates from 24+ hours to 30 minutes by using Java HttpClient and Flow APIs to streamline retrieval of polymorphic external configurations.

Firmware Engineer Intern

May 2022 – August 2022

Landis+Gyr

Alpharetta, GA

- Accelerated the process of converting DCW (Data Control Words) hexadecimal values to their corresponding ASCII representations by ~50% through efficient Python practices.
- Bolstered data transmission efficiency by ~25%, collaborating with a software architect to conduct comprehensive testing of collectors and debugging of various Python scripts employing MySQL.

Projects

Fintech @ Georgia Tech | React, React Native, TypeScript, Stripe API, Chakra UI

- Reduced page load times by ~35% via transforming a multi-page credit card website into a React-based single-page application, streamlining card wallet management and adding a card removal feature for authenticated users.
- Implemented a cart page for a React Native grocery application, allowing users to add/remove items, adjust quantities, view real-time total costs, and integrated the Stripe API for streamlined in-app credit card payments.

CDC Data Reconciliation | Python, FastAPI, SQLite, React, Tailwind CSS

- Led a team of 6 to automate the reconciliation of case counts for infectious diseases between 50 state health departments and the CDC, alongside doubling as a full-stack software developer.
- Utilized Python to compare state health department and CDC data, highlighting any discrepancies between the two datasets, saving 5000+ hours of manual reconciliation annually.

Skills

Programming Languages: Java, Python, JavaScript, TypeScript, SQL

Technologies/Frameworks: Kubernetes, GCP, Terraform, Cassandra, Elasticsearch, BigQuery, Google Firebase, Git, React, React Native, Node.js, Next.js, Express.js, SQLite, Postman, Docker, JUnit, Jest, Prometheus, PromQL

Concepts: Object-Oriented Programming & Design, Test-Driven Development, NoSQL, Data Structures & Algorithms, Full-Stack, CI/CD, Infrastructure, Distributed Systems, Cloud Computing, Microservices Architecture